AUROVILLE ECO-SERVICE ERP

Project Governance, Methodology & Decision Framework

Initiative Sponsor / Technical Developer: Dmitry Lab

Document Status: Final Version

Target Recipient: Auroville Eco-Service Administration & Governance

Language: English

Document Purpose: Definition of responsibilities, methodological boundaries, and decision-making framework

1. Project Governance Principle

Core autonomy boundaries, Open Source licensing, and scope of the free-of-charge contribution.

Dmitry Lab provides the existing digital architecture, software development, and technical documentation as an Open Source contribution. Auroville Eco-Service independently determines whether and how to review, adapt, implement, and operate the system. Consultative and technical support may be discussed separately, without creating an ongoing obligation.

1.1 Open Source Contribution & No-Obligation Framework

- **Open Source Codebase:** The existing project is available under the MIT License: `https://github.com/dmitry-lab-git/projects/tree/main/eco_service/LICENSE`
- **Installation & Deployment Guide:** Setup instructions and technical documentation are maintained in the repository README: `https://github.com/dmitry-lab-git/projects/blob/main/eco_service/readme.md`
- **Nature of Contribution:** Dmitry Lab provides the existing technical development and materials as a free-of-charge Open Source contribution. These include SQL DDL schema definitions, Python backend code, Reflex web application source, Docker deployment materials, simulation utilities, and technical documentation.
- **Development Status:** The contribution represents an existing architectural and engineering foundation for a potential ERP system. It is not a complete, production-ready ERP deployment.
- **Independent Use:** Auroville Eco-Service may independently review, use, modify, adapt, host, or deploy the materials in accordance with the applicable Open Source license and the licensing terms of any third-party components.
- **No Additional Obligation:** Receipt of the contribution does not create a separate contractual, financial, implementation, maintenance, procurement, or operational obligation for either party. Use of the software remains subject to the applicable license terms.
- **No Warranty:** The materials are provided in their existing development state, without a guarantee of fitness for a particular operational environment or production deployment. Any assessment, adaptation, security review, and validation required before operational use remain necessary.

1.2 Scope of the Existing Contribution

The contribution covers the existing repository and documentation at the time of transfer. It does not imply an obligation for continuous development, repository maintenance, production deployment, hosting, or ongoing technical support.

2. Responsibility Framework

Clear delineation between the delivered technical foundation and independent operational governance.

The following matrix distinguishes the existing technical contribution from potential future operational activities.

Functional Area	Dmitry Lab	Auroville Eco-Service
System Architecture & Design	Existing database architecture, event model, and technical design.	Review and determine operational requirements and suitability.

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Software Development	Existing SQL, Python backend, Reflex application, and simulation tools.	No software development obligation. May independently modify or extend the Open Source codebase.
Open Source & Documentation	Provides the existing repository and documentation.	Review, use, and adapt the materials independently.
Legal & Administrative Approvals	Consultative input if separately requested and agreed.	Determines and obtains any necessary community, institutional, and administrative approvals.
Site Selection & Pilot Scope	Technical guidance may be discussed separately.	Determines whether a pilot is appropriate and, if approved, selects and prepares participating sites.
Infrastructure & Hosting	Existing deployment instructions and technical guidance.	Selects, procures, configures, and administers any hosting infrastructure, domain, and backups.
Hardware & Consumables	Technical recommendations may be provided.	Determines specifications and procures any required devices, printers, smartphones, labels, and consumables.
Staff Training & Operations	Technical training or consultation may be discussed separately.	Assigns personnel, establishes operational procedures, and manages daily activities and data entry.
Operational Data & Security	Technical guidance may be discussed separately.	Establishes access permissions, security procedures, backups, and governance of operational data.

Responsibility Boundary: The matrix describes the existing contribution and the parties' respective areas of autonomy. It does not constitute an agreement to undertake future work or provide continuing services.

Operational data generated through any future implementation remains subject to the governance and ownership arrangements established by Auroville Eco-Service.

3. Methodological Distinction: Software Simulation vs. Field Pilot

Strict separation between synthetic software testing utilities and live operational field evaluation.

3.1 Software Test Data Simulation (`environment_initializer.py`)

- **Classification:** Internal software development and testing utility.
- **Purpose:** Generates synthetic data for testing database operations, event logging, application workflows, and analytical views.
- **Responsibility:** Developed and executed by Dmitry Lab as part of the existing software development process.
- **Status:** Software-level testing only. Generated records do not represent actual Auroville Eco-Service operations, measurements, or field performance.

> Synthetic data must not be presented as operational pilot statistics or used as evidence of actual waste volumes, staffing requirements, or financial performance.

3.2 Proposed ERP Field Pilot

- **Classification:** Potential subsequent operational activity.
- **Purpose:** Evaluation of the ERP system within actual waste collection, receipt, weighing, sorting, storage, and transfer workflows.
- **Decision Authority:** Auroville Eco-Service independently determines whether a field pilot should be initiated.
- **Organization & Execution:** If approved, site selection, personnel, operational arrangements, data collection, and field execution remain under Auroville Eco-Service's authority.

> **Current Status:** No field pilot has been approved or commenced. Any pilot scope, schedule, participating sites, or performance criteria remain subject to separate confirmation by Eco-Service management.

Methodological Rule: A software simulation and a field pilot are separate activities and must not be treated as interchangeable stages of the same test.

4. Decision Framework & Pending Confirmations

Illustrative four-stage evaluation sequence for independent consideration by Auroville Eco-Service.

The following sequence is proposed for consideration by Auroville Eco-Service. It is not an implementation commitment or mandatory action plan.

Stage 1 — Review of Existing Materials

- ☐ Review the Open Source repository: `https://github.com/dmitry-lab/projects/tree/main/eco_service`
- ☐ Review the installation and deployment guide (README).
- ☐ Examine the architecture, available functionality, and current development status.
- ☐ Assess the proposed system against existing operational requirements.

Stage 2 — Independent Evaluation

- ☐ Determine whether further technical evaluation is appropriate.
- ☐ Identify relevant operational requirements and potential limitations.
- ☐ Decide whether a field pilot would provide sufficient practical value.

Stage 3 — Potential Pilot Definition

Applicable only if Eco-Service decides to proceed.

- ☐ Select and approve pilot sites and participating entities.
- ☐ Define the scope, duration, and evaluation criteria.
- ☐ Select an appropriate automation level (Levels 1–5).
- ☐ Determine personnel and operational responsibilities.

Stage 4 — Infrastructure & Implementation Decisions

Applicable only after the necessary internal approvals.

- ☐ Select the preferred hosting and deployment arrangement.
- ☐ Determine hardware, printing, and consumables requirements.
- ☐ Approve any required procurement and infrastructure expenditure.
- ☐ Establish data governance, access permissions, security, and backup procedures.

All decisions, procurement, and operational commitments remain under the independent authority of Auroville Eco-Service.

5. Future Cooperation & Governance Principle

Terms of optional technical consultation and final institutional autonomy statement.

Future Cooperation: The existing Open Source contribution does not include an obligation for continuous development, production deployment, hosting, maintenance, technical support, or operational management by Dmitry Lab.

If Auroville Eco-Service identifies a need for additional development, technical consultation, training, or implementation assistance, such activities may be discussed separately. Any further cooperation would require mutual agreement on its scope and conditions.

GOVERNANCE PRINCIPLE

The technical materials are provided for independent review and potential use. Auroville Eco-Service retains autonomy over whether, when, and how the proposed system is adopted, subject to applicable licensing terms and its own institutional requirements.

THE DECISION REMAINS WITH AUROVILLE ECO-SERVICE.

End of Project Governance, Methodology & Decision Framework.